

CAPSTONE PROJECT 2

Enterprise HR & Employee Lifecycle Multi-Agent System

AI-Powered HR Operations Platform using Azure AI Foundry



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Intelligent • Human-Centric • Automated • Secure • Scalable



PROJECT OVERVIEW

An intelligent, multi-agent HR platform that automates employee lifecycle processes, delivers personalized support, and drives efficiency across the organization.

- Automate HR operations
- Enhance employee experience
- Integrate enterprise systems
- Deliver real-time insights
- Ensure security, compliance & governance

KEY CAPABILITIES

- AI-Powered HR Assistant (Conversational)
- Enterprise RAG with Policies & Documents
- Process Automation with Power Automate
- Seamless Integration with HR Systems
- Advanced Analytics & Dashboards
- Secure, Compliant & Responsible AI

ENTERPRISE INTEGRATIONS

Workday | SAP SuccessFactors | ServiceNow HRSD | Microsoft 365 (Teams, Outlook) | SharePoint | Active Directory | Power Automate

BUSINESS IMPACT

- Faster Onboarding
- Improved Employee Experience
- Reduced HR Workload
- Data-Driven Decision Making
- Scalable & Future Ready

MULTI-AGENT ECOSYSTEM





1. Executive Summary

Large enterprises manage thousands of employees across geographies, departments, and compliance frameworks. HR teams face increasing pressure to streamline:

- Employee onboarding
- Leave management
- Payroll queries
- Policy interpretation
- Employee support cases
- Lifecycle events (promotion, transfer, exit)

Traditional HR systems operate in silos, creating delays, manual workloads, and inconsistent employee experiences.

This project proposes a **Multi-Agent AI-Powered HR Operations Platform** built using modern Azure AI technologies to automate and orchestrate the complete employee lifecycle.

2. Problem Statement

Large organizations struggle to efficiently manage:

- Complex onboarding workflows
- Leave and attendance management
- Payroll-related employee queries
- HR policy retrieval
- Employee grievance handling
- Lifecycle transitions (promotion, transfer, exit)

Manual HR operations result in:

- Delays in onboarding
- Increased HR operational workload
- Poor employee satisfaction
- Compliance risks
- Inefficient cross-department coordination

The enterprise requires an **intelligent, scalable, AI-driven HR system** capable of autonomous reasoning and workflow automation.

3. Proposed Solution

AI-Powered HR Multi-Agent Platform

The solution leverages:

- Azure AI Foundry
- Azure OpenAI Service
- Azure AI Search
- Microsoft Agent Framework
- Copilot Studio
- Power Automate
- Azure Functions
- SharePoint
- SAP SuccessFactors
- Workday

4. Multi-Agent Architecture Design

The platform consists of specialized AI agents collaborating intelligently:

Agent	Responsibility
HR Assistant Agent	First-level employee support
Policy Advisor Agent	HR policy interpretation
Onboarding Agent	Automates joining formalities
Payroll Agent	Handles salary & tax queries
Lifecycle Agent	Promotion, transfer, exit workflows
Engagement Agent	Training & development insights
Compliance Agent	Policy & regulatory validation
Reporting Agent	HR analytics & dashboards

Agents coordinate through Azure OpenAI reasoning with enterprise data integration.

5. System Workflow

1. Employee interacts via Teams, Portal, or Email
2. HR Assistant Agent performs intent classification
3. Policy & HRMS data retrieved via Azure AI Search
4. Workflow triggered through Power Automate
5. Updates synchronized with Workday / SAP SuccessFactors
6. Notifications sent via Outlook / Teams
7. Dashboard updated in real time

6. Business Use Cases (Enterprise Scenarios)

Case 1: New Employee Onboarding

Solution:

- Auto-create accounts in HRMS-Database (use Agent-Query)
- Allocate laptop & assets
- Send onboarding checklist to user

Case 2: Leave Balance Inquiry

Solution:

- Integrate with Workday / Leave Tables
- Retrieve real-time leave balance
- Display upcoming leaves

Case 3: Payroll Query

Solution:

- Retrieve payroll records securely
- Provide salary breakdown
- Generate payslip copy

Case 4: Work From Home Requests

Solution:

- Refer HR policy – KB Docs
 - Trigger policy docs workflow & provide details
 - Apply Secured Guardrails & monitor regularly
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Case 5: Insurance Benefits Query

Solution:

- Retrieve policy documents from SharePoint
 - Explain coverage using summarization method to users
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Case 8: Asset Request (Laptop Allocation) from DB details

Solution:

- Validate department(table) budget
 - Trigger IT asset workflow
 - Update the system tables & provide details to users
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Case 9: Attendance (table) Correction-Query

Solution:

- Submit correction request
 - Trigger manager approval
 - Update attendance logs
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Case 10: Reimbursement Claim

Solution:

- Upload expense receipt

- Validate policy compliance
- Trigger approval & payment workflow

Case 11: HR Monthly Summary

Solution:

- Aggregate HR metrics
- Generate AI-powered analytics report
- Share dashboard with users/leadership
- Integrate into Enterprise Apps & include for multi-Agent collaboration (planner + executor + validator)

7. Security & Compliance Framework

- Role-Based Access Control (RBAC)
- Data encryption in transit & at rest
- Audit logging
- GDPR/Compliance-ready architecture
- Secure API integrations
- Confidential grievance handling

8. Performance & Business Impact

Metric	Improvement
Onboarding Time	↓ 60%
HR Manual Workload	↓ 50%
Employee Query Resolution	↑ 70% Faster
Employee Satisfaction	↑ Significant
Lifecycle Automation	↑ 65% Coverage

9. Real-Time Integrations

- Workday
 - SAP SuccessFactors
 - ServiceNow HRSD
 - SharePoint
 - Microsoft Outlook
 - Microsoft Teams
 - Payroll systems
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10. Strategic Value

This solution transforms HR from:

Manual Administration → Intelligent Digital Workforce Platform

It enables:

- End-to-end lifecycle automation
 - AI-driven employee support
 - Predictive HR analytics
 - Improved compliance governance
 - Enhanced employee experience
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11. Conclusion

The **Enterprise HR & Employee Lifecycle Multi-Agent System** demonstrates how AI-driven agent orchestration can modernize HR operations.

By combining Azure AI Foundry with enterprise HR systems, the organization achieves:

- Faster onboarding
- Reduced HR workload
- Improved employee satisfaction
- Automated lifecycle management
- Data-driven HR decision making

This platform establishes a foundation for **Autonomous HR Operations (HR-AIOps)** and future-ready workforce transformation.

Important Submission Instructions:

Deliverable Format

Participants must prepare and submit their capstone project report in either of the following formats:

- Microsoft Word Document (.docx)
- PDF Document (.pdf)

Naming Convention

Please save your file using the following naming format:

YOURNAME_EMPID.docx

OR

YOURNAME_EMPID.pdf

Example:

AjayKumar_123456.docx

or

AjayKumar_123456.pdf

Submission Guidelines

1. Ensure all sections of the capstone project are completed.
2. Include architecture diagrams, screenshots, implementation details, case solutions, and conclusions.
3. Verify that the document follows the prescribed naming convention.
4. Upload the final document to the LMS Portal folder provided by the trainer.
5. Submit before the specified deadline.

Mandatory Sections

- Cover Page
- Problem Statement

- Project Objectives
- Solution Architecture
- Tools & Technologies Used
- Implementation Details
- All Business Cases and Solutions – with Screenshots
- Security & Governance Implementation
- Monitoring & Observability Dashboard
- Performance Optimization Strategy
- Real-Time Integrations
- Conclusion
- References (if applicable)

Evaluation Criteria

- Architecture Design
- Agent Framework Implementation
- RAG Implementation
- Multi-Agent Orchestration
- Enterprise Integrations
- Security & Governance
- Monitoring & Observability
- Innovation & Business Value
- Documentation Quality
- Final Demonstration

Failure to follow the naming convention or submission guidelines may impact the evaluation process.